SMOOTHINGS OF PIECEWISE LINEAR MANIFOLDS

BY

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PREFACE

The triangulation theorems of Whitehead [12] show that to every smooth manifold V there is associated a PL manifold M and a homeomorphism $f : M \to V$, having particularly nice local properties. Moreover M and f are unique up to very strong equivalence relations. When M is triangulated, f is called a smooth triangulation of V.

Whitehead's results suggest the inverse problem: is a given combinatorial manifold M the domain of a smooth triangulation of some smooth manifold V? This is possible if and only if M has a *smoothing*, i.e., a differential structure which is compatible with its piecewise linear structure.

Smoothing theory is concerned with the problem of finding and classifying smoothings of PL manifolds. The equivalence relation we use for the classification is not the obvious one of diffeomorphisms but the stronger relation of concordance: two smoothings of M are *concordant* if they extend to a smoothing of $M \times I$. This relation has nicer functorial properties than diffeomorphism. The diffeomorphism classification of smoothings is still unknown. It turns out that concordance is the same as the relation of isotopy: two smoothings α, β of M are *isotopic* if there is a diffeomorphism $M_\alpha \to M_\beta$ which is PD isotopic to the identity as a map $M \to M_\beta$.

This book is divided into two parts. Part I is devoted to proving that every smoothing of $M \times I$ is isotopic to a product smoothing. This is a very useful stability result; it implies for example that smoothings of $M \times \mathbf{R}^k$ correspond bijectively to smoothings of M, modulo concordance. (In Part II this is extended to smoothings of vector bundles over M.) Another consequence is that concordance and isotopy are the same equivalence relation. The methods used in Part I are entirely elementary.

Part II builds on the results of Part I to classify concordance classes of smoothings of M. For each PL manifold M, there is a "PL tangent bundle" t_M over M. Such bundles are classified by homotopy classes of maps $M \to BPL$ where PL is a certain semisimplicial group. There is a natural map $BO \to BPL$. If M is smoothable then t_M has a compatible vector bundle structure which we call a linearization; and isotopic smoothings define equivalent linearizations.

The main result of Part II is that *smoothings of* M *are completely classified by linearizations of* t_M. This basic fact can be rephrased in terms of classifying maps: M is smoothable if and only if the classifying map $f_M : M \to BPL$ of t_M lifts to a map $M \to BO$; and every homotopy class of liftings can be realized by a unique concordance class of smoothings.

If M is smoothable, it turns out that concordance classes of smoothings also correspond to homotopy classes of maps $M \to PL/O$, where PL/O is the fibre of $BO \to BPL$. The set $[M, PL/O]$ of such homotopy classes has a natural abelian group structure. See in this regard [8].

The classification of smoothings has been a model for several other classification theories. For example, Kirby and Siebenmann [2,13] have shown that the problem of putting a PL structure on a topological manifold M is equivalent to putting a PL structure on the "topological tangent bundle" of M – except perhaps in dimension 4. Sullivan [3,4], Browder [1] and Novikov [9] have analogous (but more complicated) results for the problem of classifying manifolds of a given homotopy type.

The present study is entirely theoretical in that no computations are carried out. The task of actually calculating the set of concordance classes of smoothings of a given manifold is still formidable. As a consequence of Part II, however, it "reduces" to standard problems in homotopy theory. Some results of this type are contained in the references.

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Smoothings of Piecewise Linear Manifolds

SMOOTHINGS OF PIECEWISE LINEAR MANIFOLDS I: PRODUCTS

Morris W. Hirsch

§1. *Introduction*

The object of this paper is to give an elementary proof of the fundamental theorem of the theory of smoothings of PL manifolds:

MAIN THEOREM. *Let* α *be a smoothing of* $M \times I$, *inducing the smoothing* β *on* $M \times 0$; *let* $A \subset M$ *be a closed set such that* α *is a product smoothing in a neighborhood of* $A \times I$. *Given* $\varepsilon : M \times I \to \mathbf{R}_+$ *there is a diffeomorphism from* $(M \times I)_\alpha$ *to the product smoothing* $M_\beta \times I$ *which is* PD ε*-isotopic to the identity rel* $M \times 0 \cup A \times I$.

A more elaborate theorem is stated in 4.1.

Versions of this theorem have been proved by Munkres [14] and independently, the author. Both of these proofs were based on Munkres' ingenious obstruction theory. The proof presented in this paper is independent of obstruction theory, and will be used in subsequent articles as the basis for the classification of smoothings; the classification has an obstruction theory as a corollary.

The origin of the main theorem was the announcement by R. Thom [20] in 1958 that every smoothing of $M \times I$ is diffeomorphic to product. Munkres' obstruction theory [12] has Thom's theorem as an immediate consequence.

A theorem weaker than the "$M \times I$ theorem" is the "$M \times \mathbf{R}$ theorem" 7.7:

Every smoothing $(M \times \mathbf{R})_\alpha$ *is isotopic to a product* $M_\beta \times \mathbf{R}$ *and* β *is unique up to concordance.* This was proved independently by Mazur-

Poenaru [11] and the author [3]. Using the main theorem, it follows *that* β *is in fact unique up to isotopy.*

Therefore: for every $n \in Z_+$, isotopy classes of smoothings of M correspond bijectively to isotopy classes of smoothings of $M \times R^n$. This is the first step in the isotopy classification of smoothings.

§2. *Structure of the paper*

In essence our method is that of Cairns [2] (see also [23]) who used fields of transverse planes to smooth submanifolds of Euclidean space.

If α is a smoothing of $M \times I$, we would like to simultaneously smooth all the level surfaces $M \times t$ of the projection $\pi_2 : (M \times I)_\alpha \to I$ by a small isotopy. This can be done provided there exists a vector field which is everywhere transverse to the level surfaces. By uniqueness of collaring we may assume that α is a product near $M \times 0$ and $M \times 1$. A small PL isotopy allows us to assume that $(M \times I)_\alpha$ has a smooth triangulation such that π_2 is simplicially affine and *separates vertices* in $M \times (I - \partial I)$. This ensures that every point other than a vertex has a tangent vector transverse to nearby level surfaces.

The most delicate part of the proof involves a close examination of the behavior of π_2 in the star of a vertex v. Everything is essentially determined by the level surfaces of π_2 restricted to the boundary of a small smooth ball B around v. If $\pi_2(v) = y_0$ then $M \times y_0 \cap \partial B$ is a PL sphere whose dimension is $m - 1 = \dim M - 1$, and its bounds a PL m-ball D in ∂B.

We can make D a smooth submanifold of B. By applying the main theorem one dimension lower as an induction hypothesis, we can assume D is a smooth m-disk. We then make D coincide with the northern hemisphere of ∂B by a diffeotopy.

Let us identify B with the unit ball $D^{m+1} \subset R^{m+1}$. By uniqueness of collaring of S^{m-1} in $\partial D^{m+1} = S^m$, we can assume the map π_2 agrees with the projection $\pi_{m+1} : R^{m+1} \to R$ on a neighborhood E of